

Biomolecules

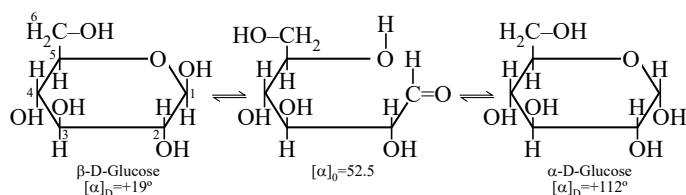
SHORT NOTES

Carbohydrates

- ❖ Carbohydrates are defined as optically active polyhydroxy aldehydes or ketones or the compound which produce such units on hydrolysis.
- ❖ **Monosaccharide** ($C_nH_{2n}O_n$): Single unit, can't be hydrolysed : Glucose and fructose.
- ❖ **Oligosaccharides** gives two to ten monosaccharides on hydrolysis.
- ❖ **Disaccharides** (by glycosidic linkage)
 - Sucrose $\xrightarrow{H_2O^+}$ α -D. Glucose + β -D. Fructose;
 - Maltose $\xrightarrow{H_2O^+}$ 2 α -D. Glucose unit
 - Lactose $\xrightarrow{H_2O^+}$ β -D. Glucose + β -D. Galactose.
- ❖ **Polysaccharide**: Contain more than ten monosaccharide units.
($C_6H_{10}O_5$)_n: Starch & cellulose.

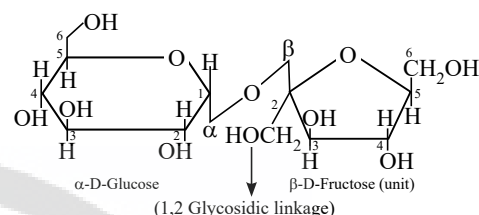
Type of Sugar		
Give Test	Reducing	Non Reducing
Tollen's Reagent	+ve test	-ve test
Fehling Reagent	+ve test	-ve test
Benedict Test	+ve test	-ve test
Mutarotation	Yes	No
Functional Unit	α -C=O / -C=O 	
Example	All monosaccharides Glucose Fructose Mannose Galactose Disaccharide Maltose Lactose	Sucrose Polysaccharide Starch Cellulose

- ❖ **Mutarotation**: When either form of D-glucose is placed in aq. solution it slowly form the other via open chain aldehyde and gradual change in specific rotation until specific rotation ($\pm 52.5^\circ$) is reached.

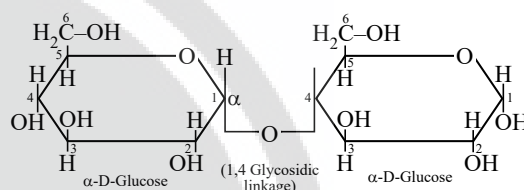


- ❖ **Anomer's**: Differ in configuration at 1st carbon due to hemi (acetal or ketal) ring formation. The new-asymmetric carbon is referred to as Anomeric carbon.
- ❖ **Epimer's**: Diastereomer's which differ in conformation at any one chiral carbon.
eg. D-Glucose & D-mannose
D-Glucose & D-Galactose

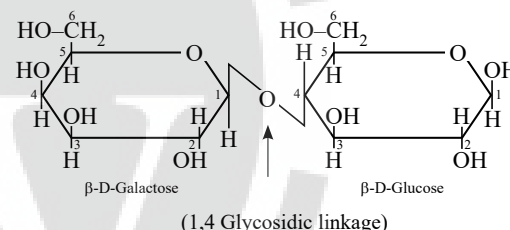
❖ Sucrose:



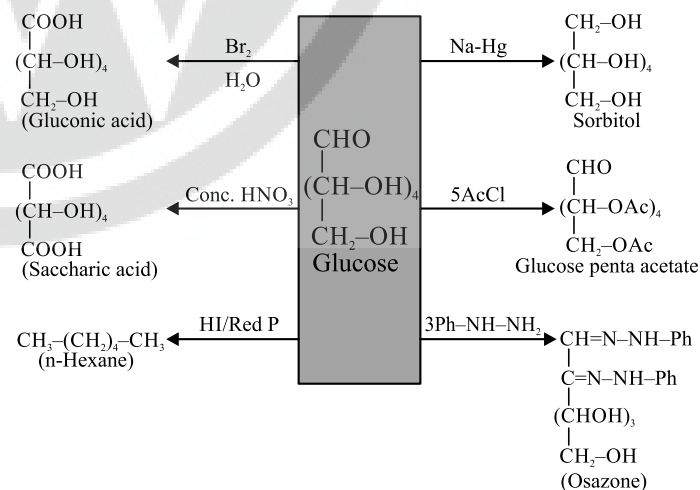
❖ Maltose:



❖ Lactose:



Reaction of Glucose (Open Chain Structure)



Amino Acids

The compounds containing amino group ($-NH_2$) and carboxylic group ($-COOH$) are called amino acids.

- (a) **Neutral amino acids**: Having one $-NH_2$ and one $-COOH$ group. E.g. glycine.

(b) **Acidic amino acids:** Having one -NH_2 and two -COOH groups. E.g. aspartic acid.

(c) **Basic amino acids:** Having two or more -NH_2 and one -COOH group. E.g. lysine.

❖ **Essential and Non-essential amino acids:** Amino acids synthesised by body are called non-essential amino acids and that are not synthesised by body are called essential amino acids.

❖ **Nomenclature:** They are known by their common names and abbreviated by first three letters of their common names e.g. glycine as 'gly' and alanine as 'ala'. Naturally occurring α -amino acids are L-amino acids.

Peptides

Peptides are condensation products of two or more amino acids, $\text{—}\overset{\text{O}}{\parallel}\text{C—NH—}$ is known as peptide linkage. A polypeptide made up of n-amino acids has (n – 1) peptide bonds.

❖ **Fibrous proteins:** Polypeptide chains are held together by hydrogen and disulphide bonds. Such proteins are insoluble in water. E.g. Keratin (present in hair, wool) etc.

❖ **Globular proteins:** These are usually soluble in water. E.g. Insulin, albumins.

❖ **Simple proteins:** These yield only α -amino acids upon hydrolysis. E.g. Albumin.

❖ **Conjugated proteins:** These yield α -amino acids and non-protein part, called prosthetic group.

❖ **Derived proteins:** These are obtained by partial hydrolysis of simple or conjugated proteins.

Denaturation of Proteins: The process that changes the three dimensional structure of native proteins is called denaturation of proteins.

Tests of Proteins

(i) **Biuret Test:** Protein solution + NaOH + dil. CuSO_4 → pink or violet colour.

(ii) **Millon's Test:** Protein solution + Millon's reagent → red colour.

Millon's reagent is a solution of mercuric nitrate and traces of sodium nitrate solution.

(iii) **Iodine reaction:** Protein solution + iodine in potassium iodide solution → yellow colour.

(iv) **Xanthoproteic test:** Protein solution + conc. HNO_3 → yellow colour $\xrightarrow{\text{NaOH}}$ orange colour.

Enzymes: Serve as biological catalyst e.g., pepsin, trypsin etc.

Vitamins: Vitamins are required by body to maintain normal health, growth and nutrition.

❖ Vitamins A, D, E and K are fat soluble whereas vitamins B and C are water soluble.

❖ Vitamin H is neither fat soluble nor water soluble.

Nucleic Acid: DNA and RNA

(i) **DNA (Deoxyribonucleic acid).**

$\text{DNA} + \text{H}_2\text{O} \rightarrow \text{Phosphoric acid} + \text{deoxyribose} + \text{A, G, C, T}$

(ii) **RNA (Ribonucleic acid).**

$\text{RNA} + \text{H}_2\text{O} \rightarrow \text{Phosphoric acid} + \text{Ribose} + \text{A, G, C, U}$

❖ In both DNA and RNA, heterocyclic base and phosphate ester linkages are at C_1 and C_5 respectively of the sugar molecule.

Hormones

These are the chemical substances which are produced by endocrine (ductless) glands in the body. Hormones acts as chemical messengers.

Insulin is a protein hormone. Its deficiency leads to diabetes mellitus.